

Security Architecture and Fault Tolerance Protocols

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1. Authentication & API Security

All REST endpoints under /mcppro-agent and /mcppro are secured via Bearer Token authorization validated in backend/app/core/auth.py. Unauthorized requests immediately abort with HTTP 401 Unauthorized.

2. Input Sanitization and Prompt Injection Defense

In the Next.js frontend, refineQuery intercepts all user prompts using a dedicated LLM pass (gpt-4o-mini) to strip HTML script tags, control sequences, and malicious search query injection modifiers.

3. Fault Handling and Graceful Fallback

- Malformed Documents: FileProcessor catches corrupted headers and unsupported extensions, returning standard error payloads rather than unhandled 500 exceptions.
- Agentic Fallback: If MasterMCPPro encounters a failure in Traditional RAG mode, it automatically triggers fallback to WorkerMCPPro agentic loop.